

EE 446 - TinyML - Lab 9: ASL Gesture Recognition

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Part 1: Software Issue Fixed

The notebook contained one intentional issue in the Build and Train the Model section (Cell 12). The model was compiled using regression metrics instead of classification metrics:

Original (incorrect) code:

```
model.compile(optimizer='rmsprop', loss='mse', metrics=['mae'])
```

This is incorrect because the model uses a softmax output layer with one-hot encoded labels, making this a multi-class classification task. Mean Squared Error (mse) and Mean Absolute Error (mae) are regression metrics and are not appropriate here.

Fixed code:

```
model.compile(optimizer='adam', loss='categorical_crossentropy', metrics=['accuracy'])
```

`categorical_crossentropy` is the correct loss function for multi-class classification with one-hot encoded labels, and `accuracy` is the appropriate metric.

Downstream Changes Made

After fixing the compile statement, the metric plotting variable in Cell 16 was also updated to match the new metric:

Original: `METRIC_TO_PLOT = "mae"`

Fixed: `METRIC_TO_PLOT = "accuracy"`

This ensures the training and validation accuracy plot titles, axis labels, and history keys are all consistent with the corrected training setup.

Part 2: Hardware Deployment - Serial Monitor Results

The trained model was converted to TensorFlow Lite format and exported as `model.h`. The `model.h` file was added to the `lab9-classifier-dual-board` Arduino sketch folder and uploaded to the Arduino Nano 33 BLE Sense. The Serial Monitor was opened to test gesture classification. The board uses the on-board IMU to detect motion and classify it as one of the two trained gestures: "hi" or "sup".

Serial Monitor - "hi" gesture correctly identified:

```
hi: 0.998054  
sup: 0.001946
```

```
hi: 0.995259  
sup: 0.004741
```

```
hi: 0.962864  
sup: 0.037136
```

```
hi: 0.384978  
sup: 0.615022
```

```
hi: 0.996632  
sup: 0.003368
```

Serial Monitor - "sup" gesture correctly identified:

```
hi: 0.204048  
sup: 0.795952
```

```
hi: 0.038007  
sup: 0.961993
```

```
hi: 0.997921  
sup: 0.002079
```

```
hi: 0.313878  
sup: 0.686122
```

Both gestures were correctly classified by the deployed model on the Arduino Nano 33 BLE Sense, confirming successful end-to-end TinyML deployment from training through on-device inference.